Curriculum Vitae - Carlos Miguel Córdoba Caycedo

Contact Information

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Education

Ph.D. in Mathematics **Sept. 2022 – May 2027 (Expected)**
McMaster University, Canada.
Advisors: Dr. Deirdre Haskell and Dr. Patrick Speissegger

Thesis: *Elimination of imaginaries in analytic and differential extensions of algebraically closed valued fields.*

B.Sc. in Mathematics **Aug. 2017 – May 2021**
Universidad de Los Andes, Colombia.
Advisors: Dr. Darío García and Dr. John Goodrick

Thesis: *Classification of Pseudofinite Abelian Groups.*

B.Sc. in Physics **Jan. 2017 – May 2021**
Universidad de Los Andes, Colombia.
Advisor: Dr. Jaime E. Forero-Romero

Thesis: *Geometrical Analysis of Cosmic Voids to Constrain Cosmological Parameters.*

Research and publications

Research Interests

Model theory (especially applications of model theory to algebraic structures), valued fields (with analytic structure), pseudofinite groups, differential algebra, o-minimality and continuous model theory.

Research Experience

Model theory research group **Sept. 2022 – Present**
McMaster University, Canada.
Weekly seminar participant. Co-organizer of the Waterloo-McMaster Joint Logic Seminar.

Visiting Researcher **Fall 2025 (August)**
Universidad de Los Andes, Colombia.

Hosted by D. Garcia and P. Cubides Kovacsics. Work on pseudofinite abelian groups, and differential analytic structure in valued fields.

Mathematical Logic Research Group **Jan. 2020 – May 2021**
Universidad de Los Andes, Colombia.
Weekly seminar participant. Work on stable and oligomorphic groups; contributed to a survey on the model theory of abelian groups.

Computational Cosmology Research Group **Aug. 2019 – May 2021**
Universidad de Los Andes, Colombia.
Weekly seminar participant. Contributed to the construction of a void-finding algorithm.

Papers and Pre-prints

Preprints (available on arXiv)

[1] **Cordoba-Caycedo, C.**, and Garcia, D. *Pseudofinite abelian groups and asymptotic classes of finite abelian groups.* arXiv:2610.13770 [math.LO], 2026.

In preparation

- Cordoba-Caycedo, C., and Haskell, D. *Elimination of imaginaries in algebraically closed valued fields with analytic and differential structure.*
- Cordoba-Caycedo, C., and Speissegger, P. *The field of reals with asymmetric multi-summable classes.*
- Hart, B., and Cordoba-Caycedo, C. *The Urysohn sphere is pseudofinite.*

Teaching

Teaching Interests

Advanced courses in differential equations, numerical methods, model theory or differential algebra. Introductory courses in analysis and calculus. Introduction to computation and data science courses.

Instructor Experience

During my career I have taught 6 courses at McMaster University, and coordinated one of them as well.

Calculus for Math and Stats I - Math 1X03 **Fall 2026 (Present)**
McMaster University, Canada.
Enrollment: 115 students.

Multivariable calculus - Math 2MC3 **Spring 2026**
McMaster University, Canada.
Enrollment: 36 students.

Numerical Methods for Differential Equations - Math 4NA3 **Winter 2026**
McMaster University, Canada.

Enrollment: 27 students. Upper level & graduate course (Math 6NA3)

Engineering Mathematics III - Math 2Z03 **Spring 2025**
McMaster University, Canada.
Coordinator of the course. Enrollment: 245 students.

Calculus for Science I - Math 1A03 **Fall 2024**
McMaster University, Canada.
Enrollment: 190 students.

Introduction to differential equations - Math 2C03 **Spring 2024**
McMaster University, Canada.
Enrollment: 30 students.

Teaching Assistantships

Leading tutorial sessions, office hours and marking for the following courses at McMaster University.

- Math 4GR3 - Groups and Rings (Winter 2026)
- Math 2C03 - Introduction to Differential Equations (Winter 2026, Fall 2025, Winter 2025, Fall 2023)
- Math 4A03 - Real Analysis II (Fall 2025)
- Math 4LT3 - Topics in logic (Fall 2025)
- Math 2XX3 - Advanced Calculus II (Winter 2024)
- Math 2ZZ3 - Engineering Mathematics IV (Winter 2023)
- Math 2Z03 - Engineering Mathematics III (Fall 2022)

Also, led the following complementary course at Universidad de Los Andes.

- Mate 1201 - Precalculus (Jan. 2021 – May 2021)

Event organization and Service

Event Organization

Waterloo–McMaster Joint Logic Seminar **Jan. 2026 – present**
University of Waterloo, and McMaster University, Canada.
Co-organizer of the joint seminar sessions in mathematical logic.

Casual Math Seminar **Sept. 2024 – Apr. 2025, Sept. 2026 – present**
McMaster University, Canada.
Member of the organizing committee of the Casual Math Seminar for both undergraduate and graduate students.

Service and others

- Treasurer, *Latin American Graduate Student Association*, McMaster University, Sept. 2026 – present.
- VP of Communications, *Mathematics & Statistics Graduate Society*, McMaster University, Jan. 2024 – Oct. 2024.
- Member of the Organization of Latin American Students (*OLAS*), and the Latin American Network at McMaster University (*LANMU*), Sept. 2022 – present.

Honours and awards

- James Stewart Award, Department of Mathematics & Statistics, McMaster University, 2025. ($1.000 CAD for travel expenses)
- Math & Sciences Graduate Scholarship, Department of Mathematics & Statistics, McMaster University, 2022-2027. (Yearly funding of $32.000 CAD)
- Laureate thesis distinction (highest distinction), Mathematics, Universidad de Los Andes, 2021.
- Laureate thesis distinction (highest distinction), Physics, Universidad de Los Andes, 2020.
- Winner, National Mathematics Contest "Supérate con el Saber", Colombia, 2014.

Lectures and presentations

Invited Conference Talks

- *Elimination of imaginaries in algebraically closed valued fields with analytic and differential structure.* XXI Latin American Symposium on Mathematical Logic (SLALM), Universidad de Los Andes, Colombia, 2026.
- *Constraining cosmological parameters using cosmic voids.* CoCo 2020: Cosmology in Colombia, 2020.

Seminar and University Talks

- *Classification of pseudofinite abelian groups and exact classes of finite abelian groups.* Model Theory seminar, McMaster University, Canada, 2025.
- *An Introduction to imaginaries in analytic extensions of algebraically closed valued fields.* Extraordinary Model Theory seminar, Universidad de Los Andes, Colombia, 2025.
- *Stable domination in a family of power series fields with analytic structure.* Model Theory seminar, McMaster University, Canada, 2024.

- *Unlimited Power: Introduction to pseudofinite structures.* Casual Math seminar, McMaster University, Canada, 2023.

Short Presentations

- *Ongoing research problems in my Ph.D. studies.* Ph.D. Student Session, Workshop on Model Theory, Algebraic Dynamics, and Differential-Algebraic Geometry, Fields Institute, Toronto, Canada, 2025.

Conferences and Workshops

Research Conferences

- *XXI Latin American Symposium on Mathematical Logic (SLALM)*, Universidad de Los Andes, Colombia, 2026.
- *XX Latin American Symposium on Mathematical Logic (SLALM)*, Universidad de la República, Uruguay, 2024.
- *Model theory and related topics,* University of Manchester, UK, 2023.
- *CoCo 2020: Cosmology in Colombia,* Virtual Conference, 2020.

Workshops

- *Workshop on Model Theory, Algebraic Dynamics, and Differential-Algebraic Geometry*, Fields Institute, Toronto, Canada, 2025.

Seasonal Schools

- *Summer School on Exponential Fields and Valued Fields in Model Theory*, University of Naples Federico II, Italy, 2025.
- *Foundations and Frontiers of Probabilistic Proofs Summer Graduate School*, École Polytechnique Fédérale de Lausanne (EPFL), Zürich, Switzerland, 2023.

Symposia

- *2025 Fields Medal Symposium: Mark Braverman*, Fields Institute, Toronto, Canada, 2025.

Professional Experience

Senior Data Quality Specialist - Math **Dec. 2024 - June 2026**
Cohere, Canada.
AI training and understanding in multiple areas of pure mathematics.

Corporate Data Analyst **May 2022 - Aug. 2022**
Lean Solutions Group, Colombia.
Leading the creation of a data science division in the company.

Big Data Analyst **Nov. 2021 - May 2022**
Lean Solutions Group, Colombia.
Leading the data analysis team for one division of the company.

Big Data Assistant **June 2021 - Nov. 2021**
Lean Solutions Group, Colombia.
Founding member of the data analysis team.

Software and languages

Software and programming languages

- Advanced experience in Python, R, C++.
- Intermediate experience with MATLAB.
- Advanced experience writing in LaTeX.
- Professional experience in Power BI, Power automate, Claude Code/Anthropic API.

Languages

- Spanish (Native)
- English (Professional proficiency)